mental operations and in an extensive trade. It was impossible to carry on without simple methods of calculation.

The adoption of zero and the decimal place-value system in India unbarred the gates of the mind to rapid progress in arithmetic and algebra. Fractions come in, and the multiplication and division of fractions; the rule of three is discovered and perfected; squares and square-roots (together with the sign of the square-root, V) J cubes and cube-roots; the minus sign; tables of sines; n is evaluated as 3-1416; letters of the alphabet are used in algebra to denote unknowns; simple and quadratic equations are considered; the mathematics of zero are investigated. Zero is defined as a — a = 0; a + 0 = a; a - 0 = a; a x 0 = 0 ; a 0 becomes infinity. The conception of negative quantities also comes in, thus: 4 = ± 2.

These and other advances in mathematics are contained in books written by a succession of eminent mathematicians from the fifth to the twelfth century **A.C**. There are earlier books also (Baudhayana, *c.* eighth century **B.C.;** Apastamba and Katyayana, both *c.* fifth century **B.C.)** which deal with geometrical problems, especially with triangles, rectangles, and squares. But the earliest